

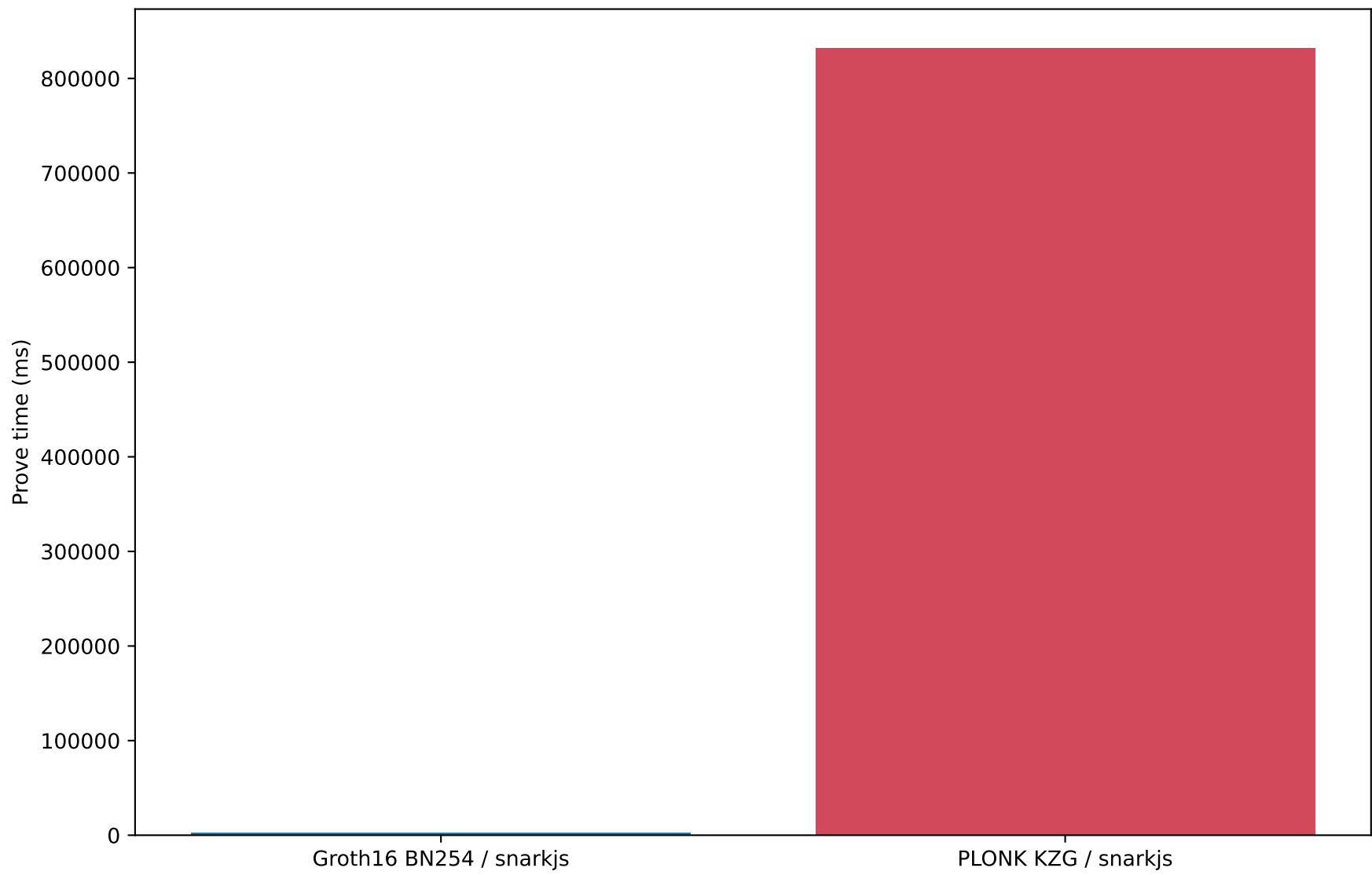
Zero-knowledge proof systems (new)

Actual Groth16 and PLONK proofs generated and verified against the same Circom R1CS statement and common witness.

System	Trial	Witness ms	Prove ms	Verify ms	JSON proof B	Compact B	RSS prove MB	Valid	Tamper rejected
Groth16 BN254 /	1	433.9	2580.6	848.4	808	129	1115.9	True	True
PLONK KZG / snark	1	422.6	831694.6	1013.6	2251	—	19819.7	True	True

Measured prove time

PLONK setup is reported separately and excluded from steady-state prove latency.



Setup, trust and scope limits

Only measured facts are plotted. Any unimplemented alternative remains unmeasured.

Scope	Details
Same statement	Same existing Circom payload_verify R1CS, same payload and 32-byte secret; proves SH
Constraints	62553
PLONK setup	668000.0 ms
Provenance	Groth16 proving artifact ceremony provenance not independently certified.
Not measured	Halo2, RISC Zero/STARK, Bulletproofs, and other systems are unimplemented; do not sub
Proof boundary	Proof concerns circuit relation, not camera origin or complete H.264 integrity.